

# Supplemental Material for “Exact Replica Mechanics and Born-Record Large Deviations in Monitored Quantum Circuits”

Amin Abaee<sup>1,\*</sup>

<sup>1</sup>*Independent Researcher*

(Dated: September 20, 2026)

This Supplemental Material collects the reference data behind Sec. II.F and Appendix A of the main text: the exceptional sets  $E_L$  of the three-replica compressed operator for  $d = 2$  and  $d = 3$  with blocks, minimal-polynomial degrees and exact ranks (Sec. SI), the per-block certificate data at  $L = 12$  (Sec. SII), the endpoint and  $p_\times$  ranks (Sec. SIII), and the complete list of certificate files and scripts with SHA-256 hashes and reproduction commands (Sec. SIV). Every number is exact (integer, modular with the stated prime, or algebraic with a certified isolating interval); no floating-point computation enters. Notation follows the main text:  $\mathbf{B}_p$  is the integer-polynomial matrix of Eq. (A1),  $\Gamma = H \times \mathbb{Z}_m$  with  $H = (S_3 \times S_3) \rtimes \mathbb{Z}_2$ , blocks are labelled  $(\rho, k)$  by an irreducible representation  $\rho$  of  $H$  (dimension  $d_\rho$ ;  $\iota\pm$  is the transposition parity, Ind an induced representation of dimension 2 or 4) and a momentum  $k$  of  $\mathbb{Z}_m$  ( $k = \pm 1$  and  $k = \pm 2$  denote the rational block that is the sum of the two conjugate complex sectors).

## SI. EXCEPTIONAL SETS

At a point  $p^* \in E_L$  the drop of rank  $\mathbf{B}_{p^*}$  below the generic rank is  $\sum d_\rho \nu_\rho$ , where  $\nu_\rho$  is the nullity of the block at  $p^*$ ; the tables list, for each  $p^*$ , the block(s) whose invariant factor vanishes, the degree of the minimal polynomial  $f$  of  $p^*$  over  $\mathbb{Q}$ , the nullity, and the exact rank of  $\mathbf{B}_{p^*}$ . The numerical value of  $p^*$  is a truncation of the certified isolating interval (all intervals have width below  $10^{-15}$ ). For  $d = 2$  every interior zero is simple (certificate multiplicity equal to the nullity in all 11 block factors at  $L = 8$  and the one at  $L = 6$ ); for  $d = 3$  all are simple except the linear factor  $3p - 2$  ( $p_\times(3) = 2/3$ ) in the  $k = \pm 1$  blocks  $(\text{triv} \otimes \text{triv})^{\iota+}$  at  $L = 6$ ,  $(\text{std} \otimes \text{std})^{\iota+}$  at  $L = 6, 8$  and  $(\text{sgn} \otimes \text{sgn})^{\iota+}$  at  $L = 8$  (file `n3_simple_zero_check.log`).

## SII. THE $L = 12$ CERTIFICATE, BLOCK BY BLOCK

Table S4 lists, for the momentum- $\pi$  sector  $k = 3$  of  $L = 12$  ( $m = 6$ ), the multiplicity  $m_{\rho,3}$ , the reflection-parity dimensions  $(m_+, m_-)$ , the symmetry-forced bound  $2 \min(m_+, m_-)$ , the numbers  $(c_+, c_-)$  of exactly verified

TABLE S1.  $E_8$  for  $d = 2$  (generic rank 1202): the eleven algebraic points in  $(0, 1)$ . For  $L = 6$ ,  $d = 2$ , the single point is  $p^* = (9 - \sqrt{41})/4 = 0.649219\dots$ , block  $\text{Ind}(\text{sgn} \otimes \text{std})$ ,  $k = \pm 1$ , nullity 2, rank 208;  $E_4 = \emptyset$ .

| $p^*$    | block $\rho$                                 | $k$     | $\deg f$ | nullity | rank |
|----------|----------------------------------------------|---------|----------|---------|------|
| 0.255038 | $\text{Ind}(\text{triv} \otimes \text{std})$ | 0       | 9        | 1       | 1198 |
| 0.270503 | $(\text{std} \otimes \text{std})^{\iota+}$   | 0       | 14       | 1       | 1198 |
| 0.381759 | $(\text{triv} \otimes \text{triv})^{\iota+}$ | 0       | 19       | 1       | 1201 |
| 0.404429 | $\text{Ind}(\text{triv} \otimes \text{std})$ | 0       | 8        | 1       | 1198 |
| 0.510263 | $(\text{std} \otimes \text{std})^{\iota-}$   | $\pm 1$ | 6        | 2       | 1194 |
| 0.548044 | $(\text{triv} \otimes \text{triv})^{\iota+}$ | $\pm 1$ | 11       | 2       | 1200 |
| 0.557093 | $\text{Ind}(\text{triv} \otimes \text{std})$ | $\pm 1$ | 12       | 2       | 1194 |
| 0.684029 | $\text{Ind}(\text{triv} \otimes \text{std})$ | 2       | 5        | 2       | 1194 |
| 0.775468 | $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | $\pm 1$ | 11       | 2       | 1200 |
| 0.862494 | $\text{Ind}(\text{triv} \otimes \text{std})$ | $\pm 1$ | 12       | 2       | 1194 |
| 0.960754 | $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | 0       | 22       | 1       | 1201 |

TABLE S2. Exceptional points for  $d = 3$  at  $L = 4$  (generic rank 21) and  $L = 6$  (generic rank 216). At  $L = 6$  the rational point  $p_\times(3) = 2/3$  affects seven blocks simultaneously— $(\text{triv} \otimes \text{triv})^{\iota+}$ ,  $k = \pm 1$  (nullity 2);  $\text{Ind}(\text{triv} \otimes \text{sgn})$ ,  $k = 0$  (1) and  $k = \pm 1$  (2);  $(\text{std} \otimes \text{std})^{\iota+}$ ,  $k = 0$  (1) and  $k = \pm 1$  (4);  $\text{Ind}(\text{triv} \otimes \text{std})$ ,  $k = 0$  (1) and  $k = \pm 1$  (2)—with total drop 40 and rank 176.

| $L$ | $p^*$    | block $\rho$                                 | $k$     | $\deg f$ | $\nu$ | drop | rank |
|-----|----------|----------------------------------------------|---------|----------|-------|------|------|
| 4   | 0.779812 | $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | 0       | 5        | 1     | 1    | 20   |
| 6   | 0.393762 | $\text{Ind}(\text{sgn} \otimes \text{std})$  | $\pm 1$ | 4        | 2     | 8    | 208  |
| 6   | 0.617321 | $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | 0       | 14       | 1     | 1    | 215  |
| 6   | 2/3      | seven blocks (caption)                       |         | 1        |       | 40   | 176  |
| 6   | 0.744876 | $(\text{triv} \otimes \text{triv})^{\iota+}$ | 0       | 16       | 1     | 1    | 215  |
| 6   | 0.775117 | $(\text{std} \otimes \text{std})^{\iota+}$   | 0       | 13       | 1     | 4    | 212  |
| 6   | 0.826750 | $(\text{std} \otimes \text{std})^{\iota+}$   | $\pm 1$ | 2        | 2     | 8    | 208  |

kernel vectors on the two parity sides, the resulting upper bound  $2 \min(m_+ - c_+, m_- - c_-)$  valid for all  $p$  outside a finite set, and the modular lower bound at  $p = 1/2$ ; the two bounds coincide in every block. The sectors  $k \neq 3$  have full multiplicity rank, 38906 in total, so the generic rank is  $38906 + 7104 = 46010 = r_3(12) - 70$ . The block data are identical for  $d = 2$  and  $d = 3$  (files `n3_L12_final.json`, `n3_L12_final_d3.json`).

Table S5 gives the modular ranks of every block  $(\rho, k)$  at  $L = 12$  for three values of  $p$ : a generic point ( $p = 1/2$ ,  $d = 2$  and  $d = 3$  give identical tables), the endpoint

\* ORCID: 0000-0002-0019-1842; Email: abaee@ut.ac.ir

TABLE S3.  $E_8$  for  $d = 3$  (generic rank 1202): 55 points. At  $p_\times(3) = 2/3$  seven blocks lose rank simultaneously— $(\text{sgn} \otimes \text{sgn})^{\epsilon+}$ ,  $k = \pm 1$  (nullity 2);  $\text{Ind}(\text{triv} \otimes \text{sgn})$ ,  $k = 0$  (1) and  $k = \pm 1$  (4);  $(\text{std} \otimes \text{std})^{\epsilon+}$ ,  $k = 0$  (1) and  $k = \pm 1$  (6);  $\text{Ind}(\text{sgn} \otimes \text{std})$ ,  $k = 0$  (1) and  $k = \pm 1$  (4)—for a total drop of 60 and rank 1142. For the two degree-46 and degree-50 factors of  $(\text{std} \otimes \text{std})^{\epsilon+}$  the nullity is pinned by  $1 \leq \nu \leq \text{mult}_f(\det)$  and the evenness of  $\nu$  in  $k = \pm 1$  blocks, as stated in Appendix A.3.

| $p^*$    | block $\rho, k$                                        | deg $f$ | $\nu$ | drop | rank | $p^*$    | block $\rho, k$                                      | deg $f$ | $\nu$ | drop | rank |
|----------|--------------------------------------------------------|---------|-------|------|------|----------|------------------------------------------------------|---------|-------|------|------|
| 0.019707 | $\text{Ind}(\text{sgn} \otimes \text{std}), 0$         | 11      | 1     | 4    | 1198 | 0.698975 | $\text{Ind}(\text{sgn} \otimes \text{std}), 0$       | 14      | 1     | 4    | 1198 |
| 0.043971 | $(\text{triv} \otimes \text{triv})^{\epsilon+}, 0$     | 49      | 1     | 1    | 1201 | 0.699932 | $(\text{std} \otimes \text{std})^{\epsilon+}, \pm 1$ | 46      | 2     | 8    | 1194 |
| 0.059680 | $(\text{triv} \otimes \text{triv})^{\epsilon+}, \pm 1$ | 25      | 2     | 2    | 1200 | 0.717603 | $\text{Ind}(\text{triv} \otimes \text{sgn}), 0$      | 6       | 1     | 2    | 1200 |
| 0.114953 | $(\text{std} \otimes \text{std})^{\epsilon-}, \pm 1$   | 10      | 2     | 8    | 1194 | 0.726117 | $(\text{sgn} \otimes \text{sgn})^{\epsilon+}, 0$     | 40      | 1     | 1    | 1201 |
| 0.116570 | $\text{Ind}(\text{triv} \otimes \text{std}), 0$        | 23      | 1     | 4    | 1198 | 0.727191 | $(\text{sgn} \otimes \text{sgn})^{\epsilon+}, 0$     | 40      | 1     | 1    | 1201 |
| 0.157295 | $(\text{std} \otimes \text{std})^{\epsilon+}, \pm 1$   | 46      | 2     | 8    | 1194 | 0.727894 | $\text{Ind}(\text{triv} \otimes \text{std}), 0$      | 12      | 1     | 4    | 1198 |
| 0.280527 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$    | 26      | 2     | 8    | 1194 | 0.734664 | $(\text{std} \otimes \text{std})^{\epsilon+}, \pm 1$ | 46      | 2     | 8    | 1194 |
| 0.327751 | $(\text{std} \otimes \text{std})^{\epsilon+}, \pm 1$   | 46      | 2     | 8    | 1194 | 0.736197 | $\text{Ind}(\text{sgn} \otimes \text{std}), 0$       | 11      | 1     | 4    | 1198 |
| 0.387717 | $\text{Ind}(\text{triv} \otimes \text{std}), 2$        | 10      | 2     | 8    | 1194 | 0.739118 | $\text{Ind}(\text{sgn} \otimes \text{std}), \pm 1$   | 28      | 2     | 8    | 1194 |
| 0.406403 | $(\text{sgn} \otimes \text{sgn})^{\epsilon+}, 0$       | 40      | 1     | 1    | 1201 | 0.747220 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$  | 26      | 2     | 8    | 1194 |
| 0.412493 | $\text{Ind}(\text{triv} \otimes \text{std}), 2$        | 10      | 2     | 8    | 1194 | 0.750329 | $(\text{std} \otimes \text{std})^{\epsilon-}, \pm 1$ | 10      | 2     | 8    | 1194 |
| 0.449885 | $(\text{std} \otimes \text{std})^{\epsilon-}, 0$       | 7       | 1     | 4    | 1198 | 0.752473 | $(\text{std} \otimes \text{std})^{\epsilon+}, 0$     | 50      | 1     | 4    | 1198 |
| 0.451387 | $(\text{std} \otimes \text{std})^{\epsilon+}, 0$       | 7       | 1     | 4    | 1198 | 0.778267 | $(\text{sgn} \otimes \text{sgn})^{\epsilon+}, 0$     | 40      | 1     | 1    | 1201 |
| 0.482738 | $(\text{std} \otimes \text{std})^{\epsilon-}, 0$       | 4       | 1     | 4    | 1198 | 0.803763 | $(\text{sgn} \otimes \text{sgn})^{\epsilon+}, 0$     | 40      | 1     | 1    | 1201 |
| 0.488163 | $(\text{sgn} \otimes \text{sgn})^{\epsilon+}, 0$       | 3       | 1     | 1    | 1201 | 0.818251 | $\text{Ind}(\text{sgn} \otimes \text{std}), 0$       | 14      | 1     | 4    | 1198 |
| 0.499209 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$    | 26      | 2     | 8    | 1194 | 0.819702 | $(\text{std} \otimes \text{std})^{\epsilon+}, \pm 1$ | 46      | 2     | 8    | 1194 |
| 0.562734 | $\text{Ind}(\text{triv} \otimes \text{std}), 0$        | 12      | 1     | 4    | 1198 | 0.826750 | $(\text{std} \otimes \text{std})^{\epsilon+}, \pm 1$ | 2       | 2     | 8    | 1194 |
| 0.604733 | $\text{Ind}(\text{triv} \otimes \text{std}), 0$        | 23      | 1     | 4    | 1198 | 0.838695 | $(\text{std} \otimes \text{std})^{\epsilon+}, 0$     | 7       | 1     | 4    | 1198 |
| 0.611851 | $\text{Ind}(\text{sgn} \otimes \text{std}), \pm 1$     | 28      | 2     | 8    | 1194 | 0.839191 | $(\text{std} \otimes \text{std})^{\epsilon+}, 0$     | 50      | 1     | 4    | 1198 |
| 0.616454 | $(\text{triv} \otimes \text{triv})^{\epsilon+}, 2$     | 14      | 2     | 2    | 1200 | 0.851727 | $\text{Ind}(\text{sgn} \otimes \text{std}), \pm 1$   | 28      | 2     | 8    | 1194 |
| 0.628181 | $(\text{triv} \otimes \text{triv})^{\epsilon+}, \pm 1$ | 25      | 2     | 2    | 1200 | 0.882507 | $(\text{std} \otimes \text{std})^{\epsilon+}, \pm 1$ | 46      | 2     | 8    | 1194 |
| 0.630557 | $(\text{std} \otimes \text{std})^{\epsilon+}, 0$       | 50      | 1     | 4    | 1198 | 0.893682 | $(\text{std} \otimes \text{std})^{\epsilon+}, 0$     | 50      | 1     | 4    | 1198 |
| 0.630590 | $(\text{triv} \otimes \text{triv})^{\epsilon+}, 0$     | 49      | 1     | 1    | 1201 | 0.895616 | $(\text{std} \otimes \text{std})^{\epsilon+}, 0$     | 7       | 1     | 4    | 1198 |
| 0.651297 | $(\text{std} \otimes \text{std})^{\epsilon+}, 0$       | 7       | 1     | 4    | 1198 | 0.942564 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$  | 26      | 2     | 8    | 1194 |
| 0.658235 | $(\text{std} \otimes \text{std})^{\epsilon+}, 0$       | 50      | 1     | 4    | 1198 | 0.951754 | $(\text{triv} \otimes \text{triv})^{\epsilon+}, 0$   | 49      | 1     | 1    | 1201 |
| 0.664027 | $(\text{std} \otimes \text{std})^{\epsilon+}, \pm 1$   | 46      | 2     | 8    | 1194 | 0.951943 | $(\text{triv} \otimes \text{triv})^{\epsilon+}, 0$   | 3       | 1     | 1    | 1201 |
| 0.694261 | $(\text{sgn} \otimes \text{sgn})^{\epsilon+}, \pm 1$   | 25      | 2     | 2    | 1200 | 0.985582 | $\text{Ind}(\text{triv} \otimes \text{std}), \pm 1$  | 26      | 2     | 8    | 1194 |

TABLE S4.  $k = 3$  sector at  $L = 12$ . Weighted totals: forced 7174, upper bound 7104, lower bound 7104.

| $\rho$                                          | $d_\rho$ | $m_{\rho,3}$ | $(m_+, m_-)$ | forced | $(c_+, c_-)$ | upper | lower |
|-------------------------------------------------|----------|--------------|--------------|--------|--------------|-------|-------|
| $(\text{triv} \otimes \text{triv})^{\epsilon+}$ | 1        | 162          | (85,77)      | 154    | (0,0)        | 154   | 154   |
| $(\text{triv} \otimes \text{triv})^{\epsilon-}$ | 1        | 55           | (19,36)      | 38     | (0,3)        | 38    | 38    |
| $(\text{sgn} \otimes \text{sgn})^{\epsilon+}$   | 1        | 170          | (99,71)      | 142    | (4,0)        | 142   | 142   |
| $(\text{sgn} \otimes \text{sgn})^{\epsilon-}$   | 1        | 61           | (30,31)      | 60     | (1,0)        | 58    | 58    |
| $\text{Ind}(\text{triv} \otimes \text{sgn})$    | 2        | 237          | (132,105)    | 210    | (4,1)        | 208   | 208   |
| $(\text{std} \otimes \text{std})^{\epsilon+}$   | 4        | 545          | (304,241)    | 482    | (5,0)        | 482   | 482   |
| $(\text{std} \otimes \text{std})^{\epsilon-}$   | 4        | 328          | (164,164)    | 328    | (1,4)        | 320   | 320   |
| $\text{Ind}(\text{triv} \otimes \text{std})$    | 4        | 411          | (210,201)    | 402    | (0,3)        | 396   | 396   |
| $\text{Ind}(\text{sgn} \otimes \text{std})$     | 4        | 423          | (234,189)    | 378    | (7,1)        | 376   | 376   |

$p = 0$  ( $d = 2$ ; identical for the second prime 1048573), and  $p_\times(3) = 2/3$  at  $d = 3$  (identical for the second prime 1048573). Blocks with  $k$  and  $-k$  have equal rank; only  $k = 0, 1, 2, 3$  are listed. Modular ranks are lower bounds

for the rank over  $\mathbb{Q}$ ; at  $p = 1/2$  they meet the upper bound of Table S4.

### SIII. ENDPOINT AND $p_\times$ RANKS

### SIV. FILES, HASHES AND REPRODUCTION

All certificate files are plain JSON or text. Table S8 lists them (63 files) with their full SHA-256 hashes; the same list, in `sha256sum` format, is the ledger `certificate_sha256.txt`, so that `sha256sum -c certificate_sha256.txt` verifies the deposit. The later deposit rounds extend the ledger without altering any entry above: `certificate_sha256_v5.txt` (the  $n = 4$  colour-resolved scans, the first  $n = 5$  diagnostics, and the Haar sequential-Monte-Carlo results quoted in the main text) and `certificate_sha256_v6.txt` (the  $n = 5$  two-size test, the  $n = 4$   $L = 10$  third crossing, and the no-

TABLE S5. Block ranks at  $L = 12$ : multiplicity  $m_{\rho,k}$  and rank at  $p = 1/2$  (generic),  $p = 0$ , and  $p = p_{\times}(3) = 2/3$  ( $d = 3$ ). Weighted totals  $\sum d_{\rho}$  rank over all  $k$ : 46010, 45822, 44898.

| block $\rho$                                 | $d_{\rho}$ | $k=0$ | $m_{\rho,k}$ |     |     | rank, $p = \frac{1}{2}$ |     |     |     | rank, $p = 0$ |     |     |     | rank, $p = \frac{2}{3}, d = 3$ |     |     |     |
|----------------------------------------------|------------|-------|--------------|-----|-----|-------------------------|-----|-----|-----|---------------|-----|-----|-----|--------------------------------|-----|-----|-----|
|                                              |            |       | 1            | 2   | 3   | 0                       | 1   | 2   | 3   | 0             | 1   | 2   | 3   | 0                              | 1   | 2   | 3   |
| $(\text{triv} \otimes \text{triv})^{\iota+}$ | 1          | 190   | 160          | 183 | 162 | 190                     | 160 | 183 | 154 | 190           | 160 | 183 | 154 | 187                            | 158 | 180 | 154 |
| $(\text{triv} \otimes \text{triv})^{\iota-}$ | 1          | 64    | 55           | 63  | 55  | 64                      | 55  | 63  | 38  | 62            | 54  | 61  | 38  | 58                             | 51  | 57  | 38  |
| $(\text{sgn} \otimes \text{sgn})^{\iota+}$   | 1          | 182   | 166          | 177 | 170 | 182                     | 166 | 177 | 142 | 178           | 163 | 172 | 142 | 171                            | 158 | 165 | 142 |
| $(\text{sgn} \otimes \text{sgn})^{\iota-}$   | 1          | 58    | 60           | 58  | 61  | 58                      | 60  | 58  | 58  | 58            | 60  | 58  | 58  | 58                             | 60  | 58  | 58  |
| $\text{Ind}(\text{triv} \otimes \text{sgn})$ | 2          | 222   | 233          | 218 | 237 | 222                     | 233 | 218 | 208 | 222           | 233 | 218 | 208 | 209                            | 223 | 205 | 206 |
| $(\text{std} \otimes \text{std})^{\iota+}$   | 4          | 560   | 545          | 560 | 545 | 560                     | 545 | 560 | 482 | 556           | 542 | 555 | 482 | 533                            | 525 | 532 | 480 |
| $(\text{std} \otimes \text{std})^{\iota-}$   | 4          | 322   | 328          | 322 | 328 | 322                     | 328 | 322 | 320 | 322           | 328 | 322 | 320 | 322                            | 328 | 322 | 320 |
| $\text{Ind}(\text{triv} \otimes \text{std})$ | 4          | 426   | 411          | 426 | 411 | 426                     | 411 | 426 | 396 | 426           | 411 | 426 | 396 | 423                            | 409 | 423 | 396 |
| $\text{Ind}(\text{sgn} \otimes \text{std})$  | 4          | 414   | 423          | 414 | 423 | 414                     | 423 | 414 | 376 | 410           | 420 | 409 | 376 | 395                            | 409 | 395 | 374 |

TABLE S6. Exact ranks of  $\mathbf{B}_p$  at the endpoints and at  $p_{\times}(d) = (d+1)/[3(d-1)]$ . Generic ranks are certified for  $d = 2, \dots, 7$  at  $L \leq 10$  and  $d = 2, 3$  at  $L = 12$ ;  $p = 0$  ranks are exact over  $\mathbb{Z}$  for  $L \leq 8$  ( $d = 2, 3$ ) and modular over two primes at  $L = 10, 12$ ;  $p_{\times}$  ranks are modular over two primes for  $d = 3, \dots, 7$  ( $L \leq 10$ ) and for  $d = 3$  at  $L = 12$  (primes 1048261 and 1048573, different seeds, identical block ranks). At  $p = 1$  the rank is 1 for every  $L$  and  $d$  (Theorem 5(iii)).

| $L$ | $6^{L/2}$ | $r_3(L)$ | generic | $p = 0$ | $p_{\times}$ | $ E_L , d = 2/3$ |
|-----|-----------|----------|---------|---------|--------------|------------------|
| 4   | 36        | 21       | 21      | 21      | 21           | 0 / 1            |
| 6   | 216       | 216      | 216     | 198     | 176          | 1 / 6            |
| 8   | 1296      | 1202     | 1202    | 1184    | 1142         | 11 / 55          |
| 10  | 7776      | 7776     | 7776    | 7663    | 7276         | ? / ?            |
| 12  | 46656     | 46080    | 46010   | 45822   | 44898        | ? / ?            |

freezing-theorem verification), each self-verifying in the same way. Table S9 lists the scripts (Python 3, dependencies `numpy` and `python-flint`; `sympy` for the  $d = 2$  number-field ranks). The reproduction sequence is given in Table S7.

The  $L = 12$  lower-bound runs take about 15 minutes each on one core with less than 1.5 GB of memory; all other steps take seconds to a few minutes.

TABLE S7. Reproduction commands (Python 3;  $\text{NB} = 2, 3, 4$  is the number of bonds,  $L = 2\text{NB}$ ).

| command                                                                   | purpose                                                       |
|---------------------------------------------------------------------------|---------------------------------------------------------------|
| <code>python3 n3_mult_certificates.py NB</code>                           | $d = 2$ interval certificates (Table II of the main text)     |
| <code>python3 n3_mult_certificates_d.py NB 3</code>                       | $d = 3$ interval certificates                                 |
| <code>python3 n3_exceptional_exact.py NB</code>                           | $d = 2$ exact ranks at the exceptional points (Table S1)      |
| <code>python3 n3_exceptional_exact_d.py NB 3</code>                       | $d = 3$ exact ranks at the exceptional points (Tables S2, S3) |
| <code>python3 n3_simple_zero_check.py</code>                              | order of vanishing of the certificate factors                 |
| <code>python3 n3_p0_exact_rank.py</code>                                  | $p = 0$ ranks over $\mathbb{Z}$ , $L \leq 8$                  |
| <code>python3 n3_closed_form_proof.py</code>                              | closed form of the Burnside deficit, Eq. (rank3)              |
| <code>python3 n3_sharpness_L12.py 6 [--d D --p P --q Q --seed S]</code>   | modular lower bounds, all blocks (Table S5)                   |
| <code>python3 n3_L12_structure_check.py</code>                            | antisymmetry of the $k = m/2$ sector                          |
| <code>python3 n3_L12_kernel.py 6</code>                                   | constant kernel vectors ( $k = 3$ )                           |
| <code>python3 n3_L12_kernel_verify.py 6</code>                            | exact verification and all- $p$ upper bound                   |
| <code>python3 n3_L12_moving_kernel.py 6 --block B --side - [--d 3]</code> | polynomial kernel vectors                                     |
| <code>python3 n3_L12_final.py [--d 3]</code>                              | two-sided certificate (Table S4)                              |
| <code>python3 n3_kernel_recheck.py [--d 3]</code>                         | independent re-verification of all kernel vectors             |

TABLE S8. Certificate files and SHA-256 hashes.

| file                                                    | SHA-256                                                           |
|---------------------------------------------------------|-------------------------------------------------------------------|
| n3_mult_cert_nb2.json                                   | 5600c82b82c7875210b9ab7803d0a5c8f6934d57f6afbc8f016679d6435ea56a  |
| n3_mult_cert_nb3.json                                   | 135635657992709144b16324a470977ed394a02bc5b21c11bc2da130294807de  |
| n3_mult_cert_nb4.json                                   | 2b618ab68941a68d53d6a0b678411f65053308c8eae8373a083aa55d155ad1dd  |
| n3_mult_cert_nb2_polys.json                             | 26fcc456ce8341f10d34c8c6abca08d04de9e0241984f7ccd43057d38e4a542a  |
| n3_mult_cert_nb3_polys.json                             | 016233445a4e8139c07dce862e1be358f8fa094df277517a942b14381b2eb837  |
| n3_mult_cert_nb4_polys.json                             | 9d944ec1dcacfb1a26216cae008fec698573cfa11de95935c084025abece505c  |
| n3_mult_cert_d3_nb2.json                                | 20cc93dfe011ab6fc20925a5310bdc3974f1aa5ab6853603c0e00e750cf89b30  |
| n3_mult_cert_d3_nb3.json                                | 7fcd62c254fff32a078dca08e268e16c254478918c6aa6c075d40228359bfe9   |
| n3_mult_cert_d3_nb4.json                                | f258be716d31e4a1d48472bfcdb364852eaeefcb8b0629733e82c4d84e7d0be78 |
| n3_mult_cert_d3_nb2_polys.json                          | 161e1ef5fc91debe27f598a6aee68199dbd35419fdc8fd4217204a5c7505973a  |
| n3_mult_cert_d3_nb3_polys.json                          | 7e23016ad43897e250fc6e089b02a84fc0eb36814d06184a8ed0c9f6ecc625d0  |
| n3_mult_cert_d3_nb4_polys.json                          | 1295e4deb729ce426eccad936240c71f7272a096c24344ccacbc6b81fd90c7    |
| n3_exceptional_exact_nb3.json                           | 55effc4a80ba193af63f449c6599bbf590e3bb0286bd015a79650b635c9996fa  |
| n3_exceptional_exact_nb4.json                           | 32442fe7d0dd851619d9a8b4c349c5595abb37df7c020b4594419f5657d66f22  |
| n3_exceptional_exact_d3_nb2.json                        | 11ce1f740cbe4d5907461c4bb730d4fd5f7e67b1d4a0a496d1c52738d119e9ee  |
| n3_exceptional_exact_d3_nb3.json                        | 785850b46a07fbde476eda6f26058fcee7e56c7ac34745a5f9e2488b2fe7390e  |
| n3_exceptional_exact_d3_nb4.json                        | e53211284b01a69f0f693296bff02d82fd5b4e42489f39012880100ab3bb8d1b  |
| n3_L12_kernel_verify_k3.json                            | 37c2581e753c771b51c84166f0decad50f9262028243d699ff7643a42b695666  |
| n3_L12_moving_kernel_2_Ind(triv.sgn)_minus.json         | 451e548b6b90b150419091f5e829d0e10a7ef9cee5f1abd56485c2386c9d98a1  |
| n3_L12_moving_kernel_4_Ind(sgn.std)_minus.json          | e685aa5e8e927203094af86049bbb093fac6f3200dd63a30046b824d2664555b  |
| n3_L12_moving_kernel_4_Ind(triv.std)_minus.json         | 9dd9a9d3cece96412693eefc9086858a03a2192402df6a442d7d0a1a7f5321a   |
| n3_L12_moving_kernel_4_std.std_i_minus.json             | 96c91b35b2fcb2905cfd0083d1217d12eba8900f09d979f1707c181943dbf55f  |
| n3_L12_moving_kernel_2_Ind(triv.sgn)_minus_d3.json      | 51cf134d4969294c65ffdaa941e2b6dc7769ef473060218a7f6ca783d9842280  |
| n3_L12_moving_kernel_4_Ind(sgn.std)_minus_d3.json       | 6eeb8cb6a8cb4f9a5734180e91eb0f720ee80a420a4ea9cdeb9e1b1f3c79abb7d |
| n3_L12_moving_kernel_4_Ind(triv.std)_minus_d3.json      | 32bc17daddb1981d9769e718825f6a92e7dae6c6c1963e71ffcac05ad0720e3d  |
| n3_L12_moving_kernel_4_std.std_i_minus_d3.json          | afe7d826252cc60b536647ad15fc9cc059494540cdbe59686a374937168527a8  |
| n3_L12_kernel_1_sgn.sgn_i+_k3_q1048261_rational.json    | 0f49fed2443e23c4f446d46cdf30befa901ba16561561bc826d25d5f127e7e7b  |
| n3_L12_kernel_1_sgn.sgn_i-_k3_q1048261_rational.json    | 4acd6bc8d9bc19682a595e82403dc175f9912d86cd6cecc5b15334dfecfc981a  |
| n3_L12_kernel_1_triv.triv_i-_k3_q1048261_rational.json  | 1d2aa2f1901995a12c92e9acf19d0f71a87617f4a73dd7128ea7a09b35fec639  |
| n3_L12_kernel_2_Ind(triv.sgn)_k3_q1048261_rational.json | 9d02b012b4f1d9b631ec1bb305be7f7fea6c6c568a8ac12b295ba4926e939285  |
| n3_L12_kernel_4_Ind(sgn.std)_k3_q1048261_rational.json  | 5652eebdda79b9a665da7ba07e77385dadb55a88df8f272404a4065da18bc530  |
| n3_L12_kernel_4_Ind(triv.std)_k3_q1048261_rational.json | 82e14c6e6396ee75589fd265d422143f39cfcc3597d463bc0026d0f8ac1fa     |
| n3_L12_kernel_4_std.std_i+_k3_q1048261_rational.json    | 33be5db62064dbdc2088fff795f25b07ef3151060ea55e3d4f23a4f5fcb6b2e3  |
| n3_L12_kernel_4_std.std_i-_k3_q1048261_rational.json    | b5edc70bc1eb3f3d47c6b110b797d7ef84435f487c272fb9044e0ff4db9d4a45  |
| n3_L12_final.json                                       | 0aa3f5a6e678b8bc80b0cba4d0a84a170a54cbdb810c9d7e666a42068c795fd75 |
| n3_L12_final_d3.json                                    | 09dcd3fc785da31010975c85b7dcec2bafef6fd8cf62230882dd0ad06dae4aff  |
| n3_sharpness_L10.log                                    | b0969da19857824e0637ee1e1a634e45039d793019161eee1333f71397d0d81f  |
| n3_sharpness_L12.log                                    | f3c2f28467d4acf15ca952587bfbfb90ebf527b4d5e39e8299002fabd7ac0930  |
| n3_sharpness_L12_k3_central.log                         | 70664316b052b71c69e6ef4ac620e8f6309c65baea54e1232d68276ace414e8   |
| n3_sharpness_L12_k3_d3.log                              | 165d9079fb4a065396d0a49d9666143be90a6a36542c3230c0b95f15888868793 |
| n3_sharpness_L12_k3_p1-2.log                            | c28642ef7b473425a9bdaee11f53440890a083c15e85a10e9ed8d91d17ed3f11  |
| n3_sharpness_L12_k3_p1-3.log                            | 568ca64a32e15eb8007b2971fb4740eab400c2ec8d73776d25fb053a7bc1d8bf  |
| n3_sharpness_L12_k3_p3-5.log                            | 129d75f4cf9e8e9a9a49ff57ee704d9850f0fec87842a087bc1693bcbb6d1b85  |
| n3_sharpness_L12_d3_full.log                            | 36060cf646724100fd3bf8af19b28c7ba8a8dcb32057ec3bbc4d7332aafa3cf6  |
| n3_sharpness_L12_p0.log                                 | 53dfefbc3ccbf3a3dd0707b942640464528b497140e1bea339bcbfd2c0dea1889 |
| n3_sharpness_L12_p0_q1048573.log                        | 1c7b9af12a9e6a3f9c4cfcb9070f73830c8dde1425dfff2e0377580f94e6bf6e  |
| n3_sharpness_L12_d3_px.log                              | 17fab566985c51be05fe068853813704e9e25fbb758ed95180cb96dd92e6aa0   |
| n3_sharpness_L12_d3_px_q1048573.log                     | 34532374c00dfa678be17c2ffe0c8cf9ae88ffd06b3585cbf307822e0efa865e  |
| n3_sharpness_L10_d3.log                                 | a1663286d77125fa8cf673084dce411df512aa9b3dd38e7fe9d4d976dbfe771d  |
| n3_sharpness_L10_d4.log                                 | af8c189121d9d15692f0add848aea65d63d96d27b52920e4ccca1c8902c9e2b7  |
| n3_sharpness_L10_d5.log                                 | a5999f67c25a9165dd946a9e5949fa185d17924e79100918e63a8530f7664234  |
| n3_sharpness_L10_d6.log                                 | 10292421bfc5316cd1fd28ca5f22d125f2fcc340773f5f7001bf03fd8e75a68b  |
| n3_sharpness_L10_d5_p1-3.log                            | cf435e2612e156df6600114189675f1853528bcbf4edf2b2281f95644f8b143e  |
| n3_sharpness_L4-10_d7_p1-3.log                          | 8b76837e6edfe453030ec5633a841661efed12a696f1d6fc36529873049096d6  |
| n3_sharpness_L4-8_d456_p1-3.log                         | 9f344ea8e8d7df8501d722fb23ecf0f3d02a2b637d17bbc2990d7850e084d18a  |

TABLE S9. Scripts and SHA-256 hashes.

| script                    | SHA-256                                                           |
|---------------------------|-------------------------------------------------------------------|
| gap_utils.py              | 784e2aa0e2275c8fa9123073a1ac267b6ca71279231801a484b7cbf410e1a3d1  |
| n3_mult_certificates.py   | ed2abc22e9dbcc091372e1845b8a9ca2fcc0cd271ec4f9c01040a37f77b9b561  |
| n3_mult_certificates_d.py | 81e8b91b0de3df0140edbc21848ccd7d42923c0b76331cc182b7e5b7f51825c   |
| n3_exceptional_exact.py   | 7b82757b018b9ac74ea963f66091eda3034c363aef842a0e74a1dfd1fceb5cadd |
| n3_exceptional_exact_d.py | 1753756feddbce19e4f630e933d0fcbe169e3a2c46776be41ac73a0a5677a791  |
| n3_simple_zero_check.py   | 30fc0dfa4795e5e637b252516f48b95c0f5f9e7dbcd4b4e703ffab2fd9d9cb35  |
| n3_p0_exact_rank.py       | adedb74bf3b3956cd4d2b77392950475a2c80da7bb449f776b3bf69c0e8ba582  |
| n3_closed_form_proof.py   | da8cc292493b9d7bc030b13a31a81b8289fd5e940beca19900b05e4abb3eb5d2  |
| n3_formula_exact.py       | 2accd8a1e5360fcb4982138b4973ed2f4848cb1cb7b270f91d85dacc144dfe3   |
| n3_sharpness_L12.py       | 3dffc312836e13489c48889d3209a862c207314f2fbc3ff6f63cebd8cf350d3   |
| n3_L12_structure_check.py | bd5064ad249d9589f5200ec9e4c5203392176391f46014f3aec46a15d8e7d312  |
| n3_L12_kernel.py          | dd962b398a388f1c6b1c26368c91a9af59c63a5d9ebc3a302f4f0e53ae581e5   |
| n3_L12_kernel_verify.py   | 548dd40a29b15c7d0fd8188f1e750aeb82fb210006054bcfed40f6fa949cad0f  |
| n3_L12_moving_kernel.py   | 07b13b7a3eac81d647290a15c86fd122bcacdc2db7ba40bea868f193550800be  |
| n3_L12_final.py           | 613584dd468066855a842c6ddcf42535bb8d428bfd9de18659422d0f742e9087  |
| n3_kernel_recheck.py      | 30062ba34504253b2d770d5bfbd795d38371ad1aeab1bafc92fd8a467c142079  |